



Management and Cost Accounting – Job & Process Costing

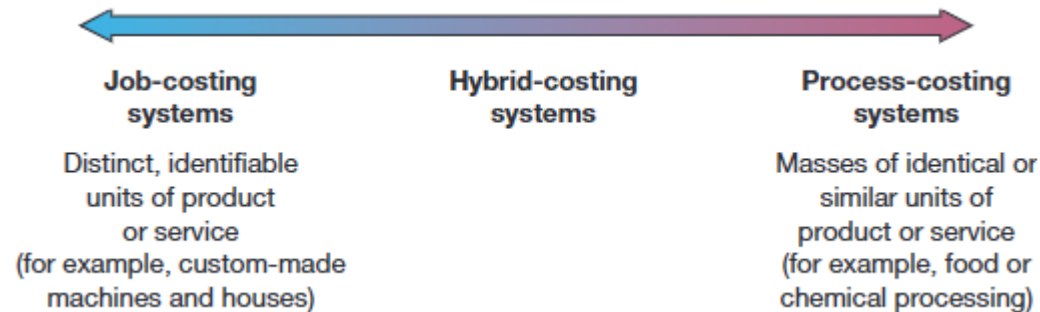
How does a job-costing system differ from a process-costing system?

Job-costing system

- Cost object = distinct unit/product/service
- Job = task for which resources are spent
- Examples: an audit job, a unique car, etc.

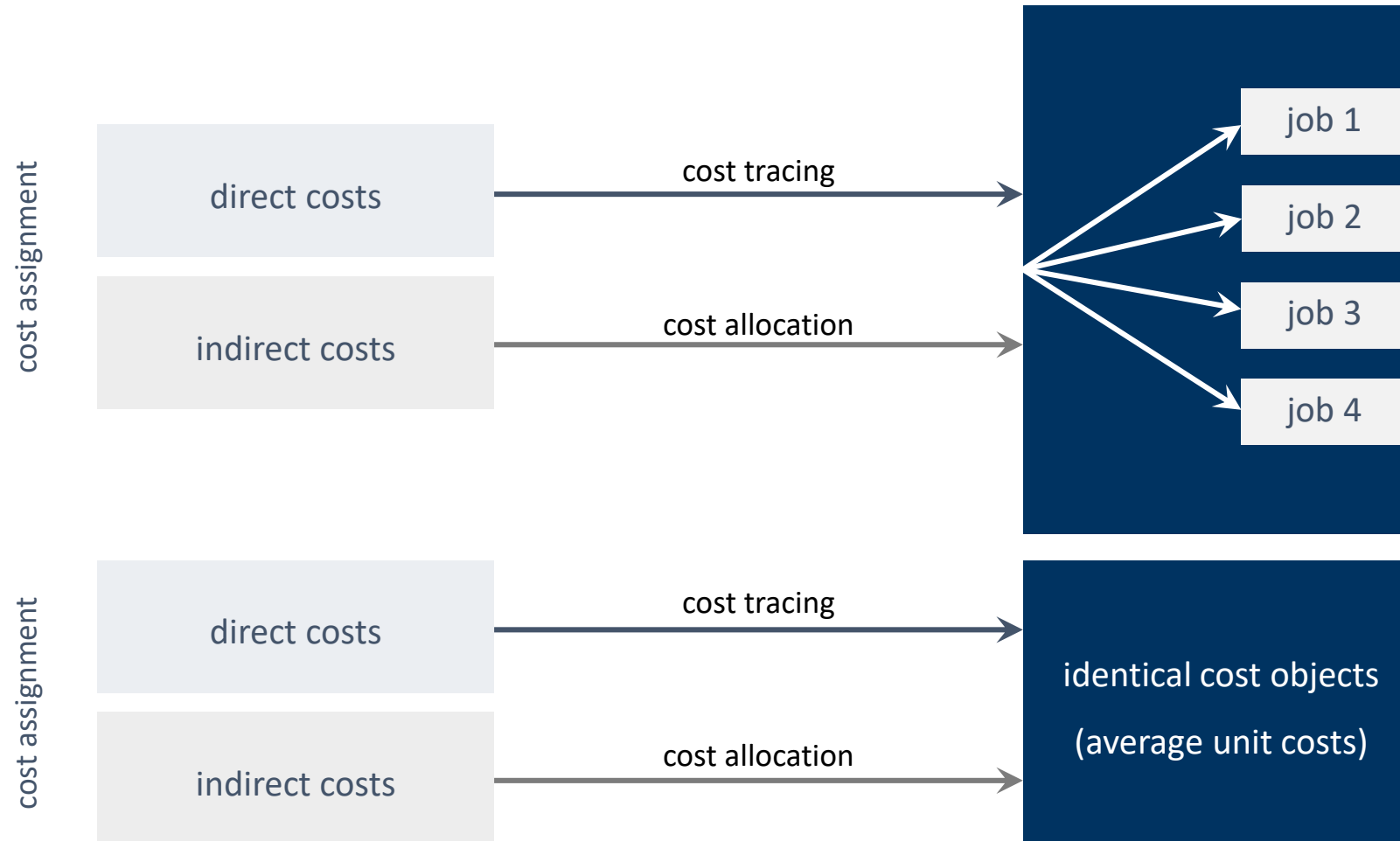
Process-costing system

- Cost object = masses of identical or similar units
- Costs are assigned to a mass of units which use resources in an identical way
- Examples: beverage production, pharmaceuticals, etc.



(BHDR, 2023, p. 116)

How does a job-costing system differ from a process-costing system?



Exercise 3.12

Zimmermann GmbH is an international consulting firm. Its annual budget includes the following for each category of professional labour:

Category	Average salary	Average fringe benefits	Billable time for clients (hours)	Vacation and sick leave (hours)	Professional development (hours)	Unbilled time due to lack of demand
Director	140,000.00€	60,000.00	1,600.00	160.00	240.00	0
Partner	105,000.00€	45,000.00	1,600.00	160.00	240.00	0
Associate	60,000.00€	20,000.00	1,600.00	160.00	240.00	0
Assistant	38,000.00€	12,000.00	1,600.00	160.00	240.00	0

- (1) Calculate the budgeted direct-cost rate for professional labour (salary and fringe benefits) per hour for (a) directors, (b) partners, (c) associates, and (d) assistants. Use budgeted billable time for clients as the denominator in these computations.
- (2) Repeat requirement 1. Use the sum of budgeted billable time, vacation and sick leave time and professional development time as the denominator in these calculations.
- (3) Why are the rates different between requirements 1 and 2? How might these differences affect job costing by Zimmermann GmbH?

(BHDR, 2023, p. 83)

Solution 3.12

Task 1 & 2)		(a) Average Salary + Average fringe benefits	(b) Billable time for clients	(c) Total time	(d) = (a) ÷ (b)	(e) = (a) ÷ (c)
	Director	200.000,00 €	1,600.00	2,000.00	125 €	100 €
	Partner	150.000,00 €	1,600.00	2,000.00	93.75 €	75 €
	Associate	80.000,00 €	1,600.00	2,000.00	50 €	40 €
	Assistant	50.000,00 €	1,600.00	2,000.00	31.25 €	25 €

Task 3)

The difference between requirements 1 and 2 is the denominator. Requirement (1) uses only billable time, leading to a **higher** cost per hour. This is an effective billing rate. Requirement (2) uses billable time + vacation/sick leave + prof. development time, leading to a **lower** cost per hour. In short, the rates are different because req. (2) spreads the same total cost over a larger total number of paid hours.

High rate (req. 1): Ensures all employee costs are covered immediately.

Low rate (req. 2): Makes the job cost less expensive, but the company has to find another way to cover the cost of paid time off and prof. training.

Exercise 3.13

Consider the following selected cost data for Schwarzmatal GmbH for 2023:

Budgeted manufacturing overhead	7,000,000.00 €
Budgeted machine-hours	200,000.00
Actual manufacturing overhead	6,800,000.00 €
Actual machine-hours	195,000.00

Schwarzmatal's job-costing system has a single manufacturing overhead cost pool (allocated using a budgeted rate based on actual machine-hours). Any amount of under- or overallocation is immediately written off to Cost of Goods Sold.

Required:

- (1) Calculate the budgeted manufacturing overhead rate.
- (2) Calculate the amount of under- or overallocation of manufacturing overhead. Is the amount significant?

(BHDR, 2023, pp. 83-84)

Solution 3.13

Task 1) Budgeted manufacturing overhead rate = $\frac{\text{budgeted overhead}}{\text{budgeted machine hours}} = \frac{€7,000,000.00}{200,000} = €35.00$ per machine hour

Task 2) Manufacturing overhead allocated = *Budgeted overhead rate x actual machine hours* = €35.00 x 195,000 = 6,825,000.00 €

€6,825,000.00 - €6,800,000.00 = **€25,000 overallocated** - an insignificant amount

Underallocated indirect costs occur when the allocated amount of indirect costs in an accounting period is **less** than the actual amount in that period.

Overallocated indirect costs occur when the allocated amount of indirect costs in an accounting period **exceeds** the actual amount in that period.

Equivalent Units

- **Equivalent units** is a derived amount of output units that takes the quantity of each input in units completed or in work in progress and converts it into the amount of completed output units that could be made with the quantity of input.
- Put differently: It is a way to measure how much work has been done on partially completed products as if they were fully finished.

(BHDR, 2023, po. 94-95)

A factory manufactures chairs:

- 100 chairs, 100% finished
- 50 chairs, 50% finished (they have the frame but no legs or cushions)
- 20 chairs, 25% finished (just the frame started)

170 chairs started

Calculation of EU:

Finished chairs: 100 EU

50% complete chairs: $50 \cdot 50\% = 25$ EU

25% complete chairs: $20 \cdot 25\% = 5$ EU

130 EU

Interpretation:

This means the workers have done the same amount of work as if they had finished 130 chairs.

Equivalent Units

Measuring Partially Completed Work as Full Units

Concept:

Converting partially finished work into full unit equivalents.

How much work has been done as if the products were fully completed.



Example: Chair Production

Factory Output:

-  100 chairs, 100% finished
-  50 chairs, 50% finished
-  20 chairs, 25% finished



Calculation of Equivalent Units

- Finished Chairs: 100 EU
- 50% Complete: $50 \times 50\% = 25$ EU
- 25% Complete: $20 \times 25\% = 5$ EU

= 130 EU

= Total Equivalent Units: 130 EU

Exercise 4.13

Consider the following data for the satellite Assembly Division of Aéro-France:

	Physical units (satellites)	Direct materials	Conversion costs
Opening work in progress (1 May)*	8	4,968,000.00 €	928,000.00 €
Started in May	50		
Completed during May	46		
Closing work in progress (31 May)†	12		
Costs added during May		32,200,000.00 €	13,920,000.00 €

*Degree of completion: direct materials, 90%; conversion costs, 40%.

†Degree of completion: direct materials, 60%; conversion costs, 30%.

Required:

- (1) Calculate equivalent units of work done in the current period for direct materials and conversion costs. Show physical units in the first column.
- (2) Calculate cost per equivalent unit of opening work in progress and of work done in the current period for direct materials and conversion costs.

(BHDR, 2023, p. 122)

Solution 4.13 (I)

Task 1)

Flow of production	(Step 1) Physical units	(Step 2) Equivalent units	
		Direct materials	Conversion costs
Completed and transferred out during current period	46	46	46
Add work in progress closing [12 * 60%; 12 * 30%]	12	<u>7.2</u>	<u>3.6</u>
Total accounted for	58	53.2	49.6
Deduct work in progress, opening [8 * 90%; 8 * 40%]	8	<u>7.2</u>	<u>3.2</u>
Started during current period	50		
Work done in current period only		46	46,4

The primary goal of the **work done in current period only** line is to measure *only* the productive effort added during the current period (= month of May). We isolate the costs that belong exclusively to the current period (= May).

We add **work in progress closing** because these 12 units are still in the factory at the end of May. They were **partially worked** on *during* May.

The **opening work in process** was already partially completed in the *previous month* (April). By **deducting WIP** opening we are essentially removing the effort from the previous month.

Solution 4.13 (II)

Task 2)

Direct materials

Conversion costs

Equivalent unit costs of opening work in progress

Work in progress, opening	4,968,000.00 €	928,000.00 €
EU of opening work in progress	7,2	3,6
Cost/EU of opening work in progress	690,000.00 €	290,000.00 €

Equivalent unit costs of work done in current period

Costs added in current period	32,200,000.00 €	13,920,000.00 €
EU of work done in current period	46	46.4
Cost/EU of work done in current period	700,000.00 €	300,000.00 €

Questions?



<https://www.pinterest.de/pin/799037158920259827/>

References

Bhimani, A., Horngren, C. T., Datar, S. M., & Rajan, M. V. (2023). *Management and Cost Accounting* (8th ed.). Pearson.